Plots for Short Dataset

Gyroscope data (X,Y,Z)

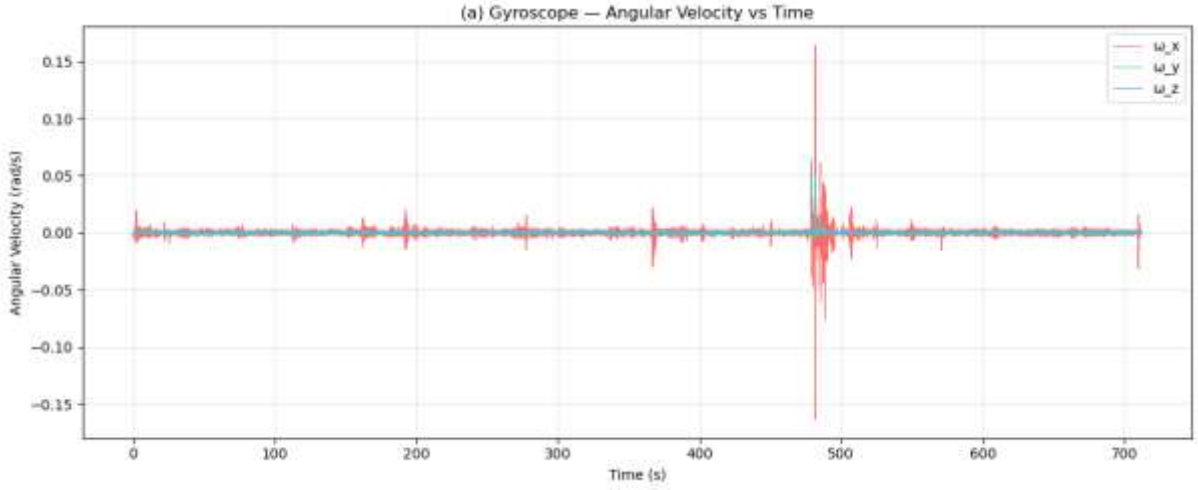


Fig1: Angular rate (X,Y,Z) vs time of stationary IMU

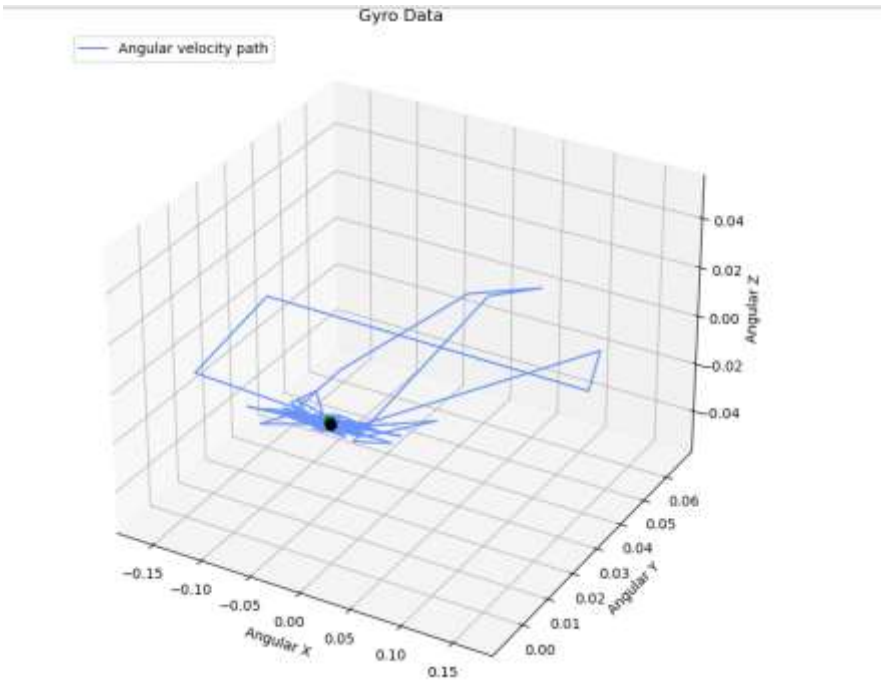


Fig2: Angular rate (X,Y,Z) data of stationary IMU

Acceleration data(X,Y,Z)

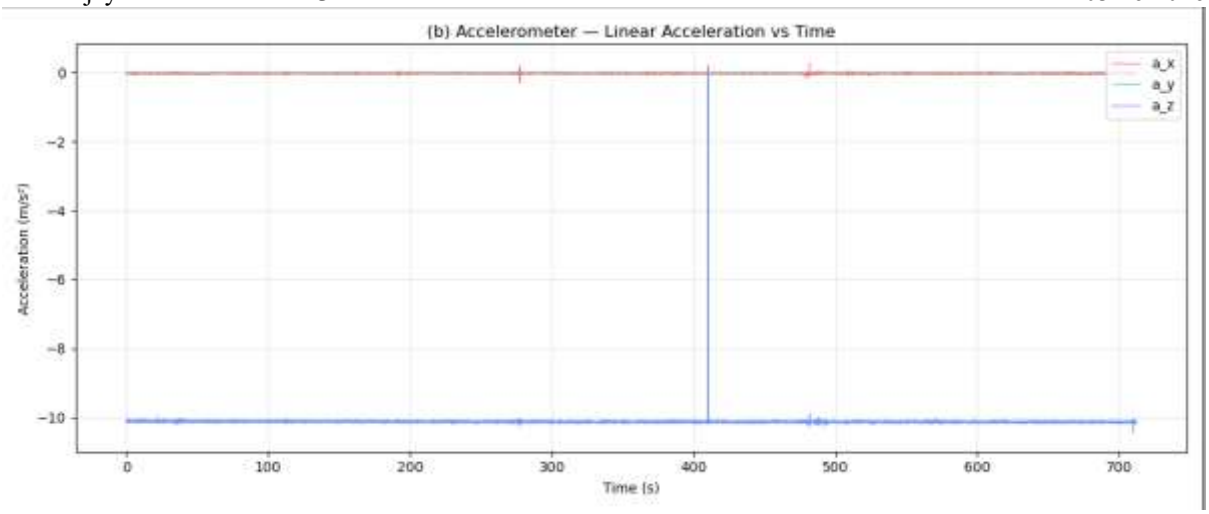


Fig3: Acceleration (X,Y,Z) vs time of stationary IMU

Magnetometer data(X,Y,Z)

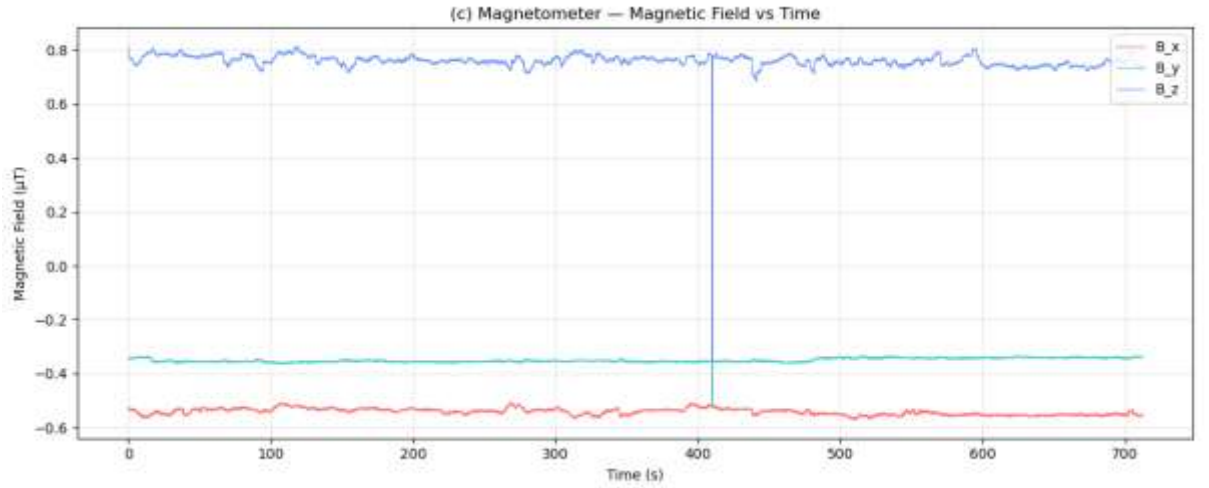


Fig4: Magnetic Field (X,Y,Z) vs time of stationary IMU

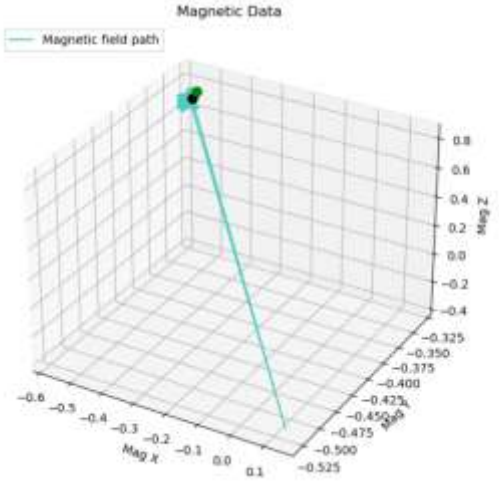


Fig5: Magnetic Field(X,Y,Z) data of stationary IMU

Orientation data(Roll, Pitch, Yaw)

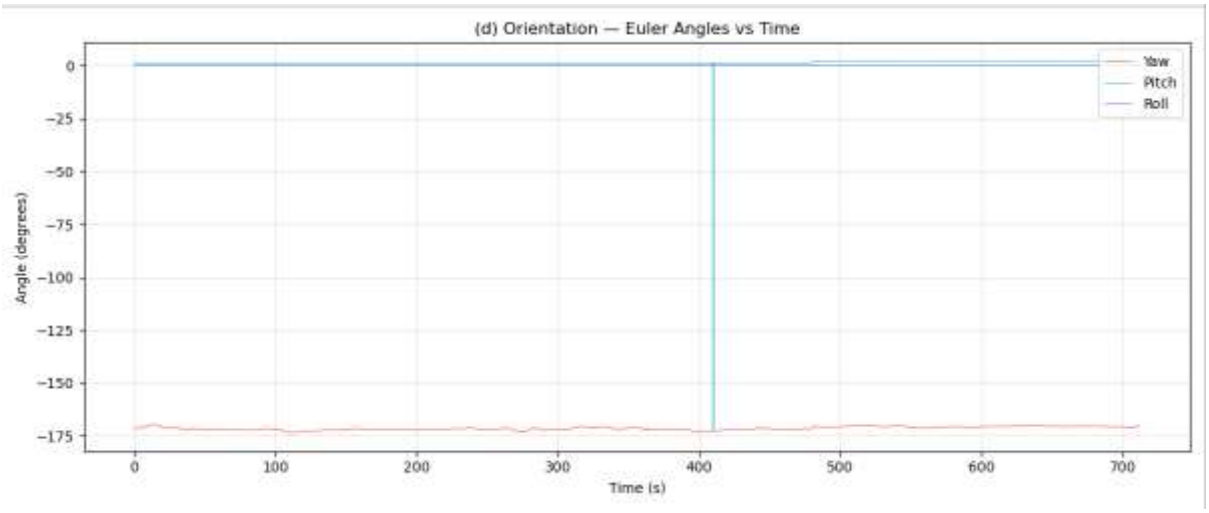


Fig6: Euler angles (X,Y,Z) vs time of stationary IMU

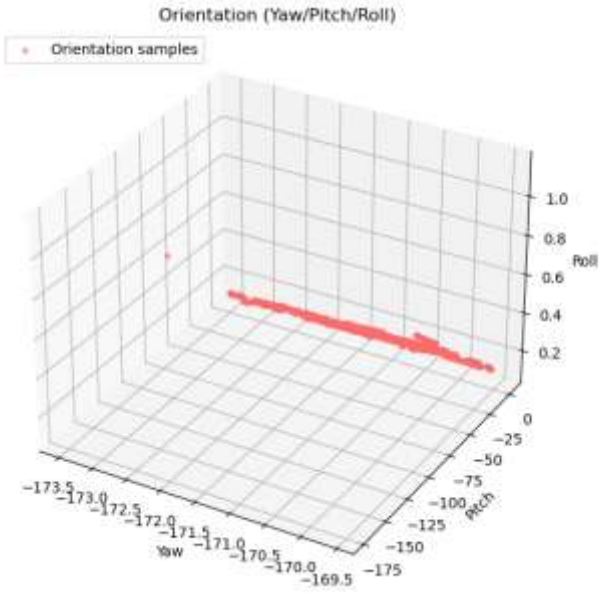


Fig7: Euler Angles (X,Y,Z) data of stationary IMU

Histogram data:

Histograms about median:

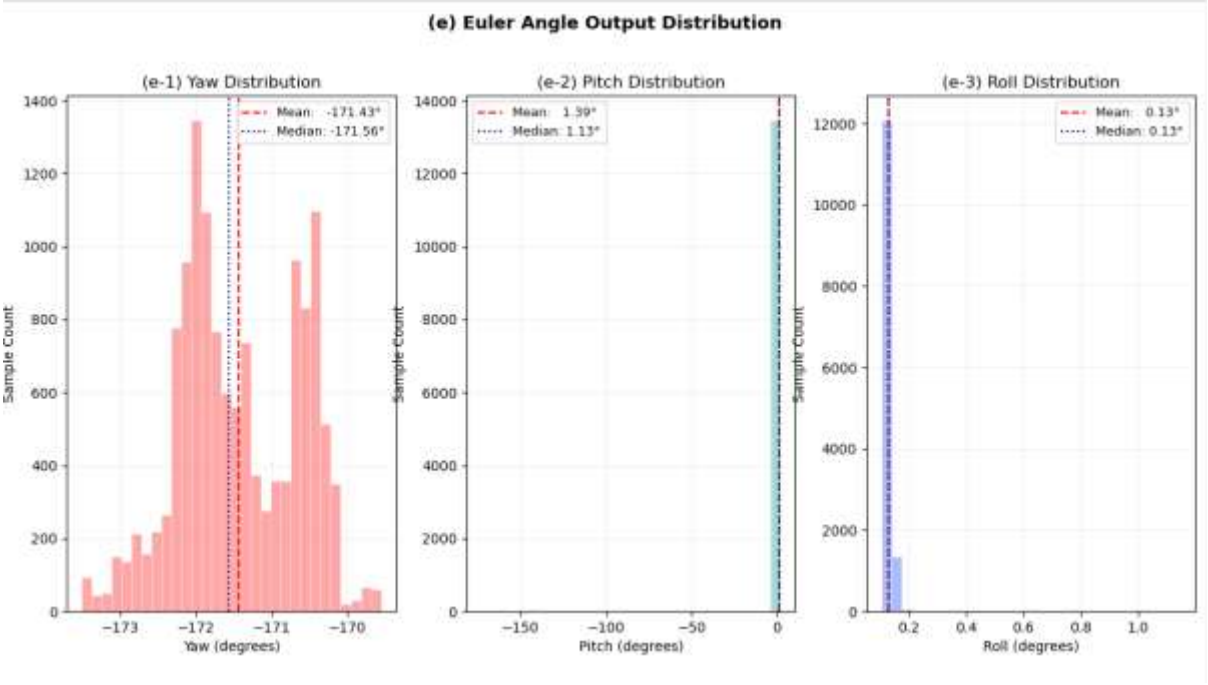


Fig8: Output distributions of Euler Angles (Yaw, Pitch, Roll) of stationary IMU dataset

The sensor was placed in a stationary position with a certain angle about z-axis (almost -170 degrees) and hence, we get the yaw to be in the range we see above.

Mean Yaw = -171.43 degrees

Median Yaw = -171.56 degrees

Mean Pitch = 1.39 degrees

Median Pitch = 1.13 degrees

Mean Roll = 0.13 degrees

Median Roll = 0.13 degrees

From this data I can conclude that, the IMU is more sensitive to Yaw and less sensitive to Pitch and Roll.

Yaw distribution: Roughly bimodal- two peaks observed around -172 and -170.4 and no peak around the mean with a spread of ~3 degrees .

Roll distribution: Single peak distribution with spread of about 0.2 degrees with mean and median at 0.13 degrees.

Pitch distribution: Single peak observed around the mean 1.39 degree, very certain/ tight distribution.

Motion Analysis:

Our Team recorded the data for about 2-3 minutes amongst which this 70s were the ones which is where the motion happened actually, where in the message count was 1428 samples. **The intended motion was acceleration in X direction, acceleration in Y direction and rotating it about Z-axis.**

The entire motion dataset can be plotted as,

Orientation

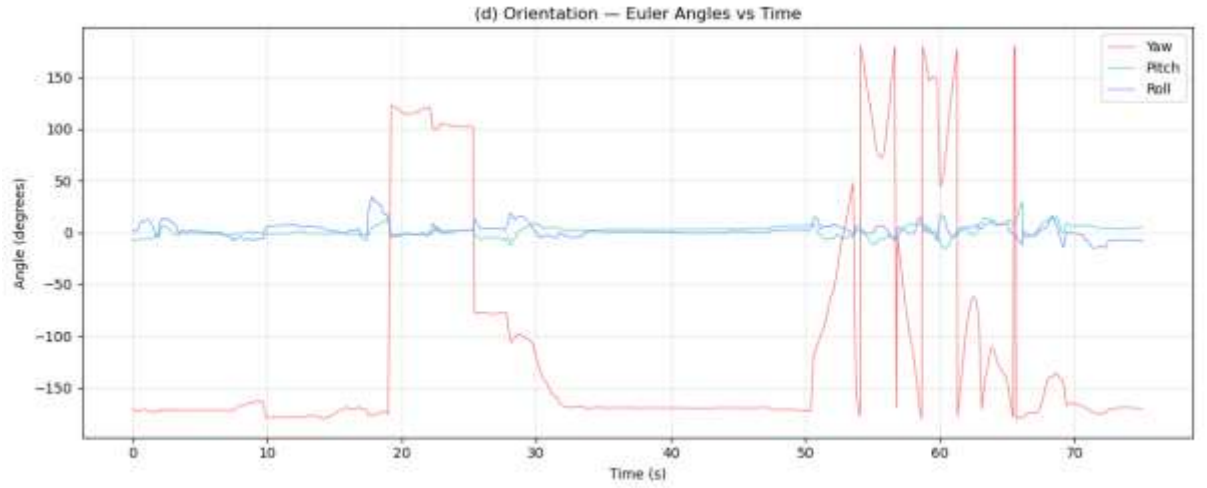


Fig9: Euler Angles(X,Y,Z) vs time of moving IMU dataset

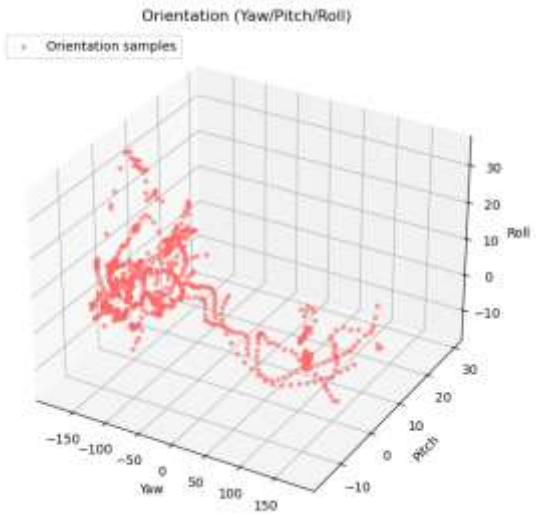


Fig9: Euler Angles(X,Y,Z) of moving IMU dataset

Acceleration

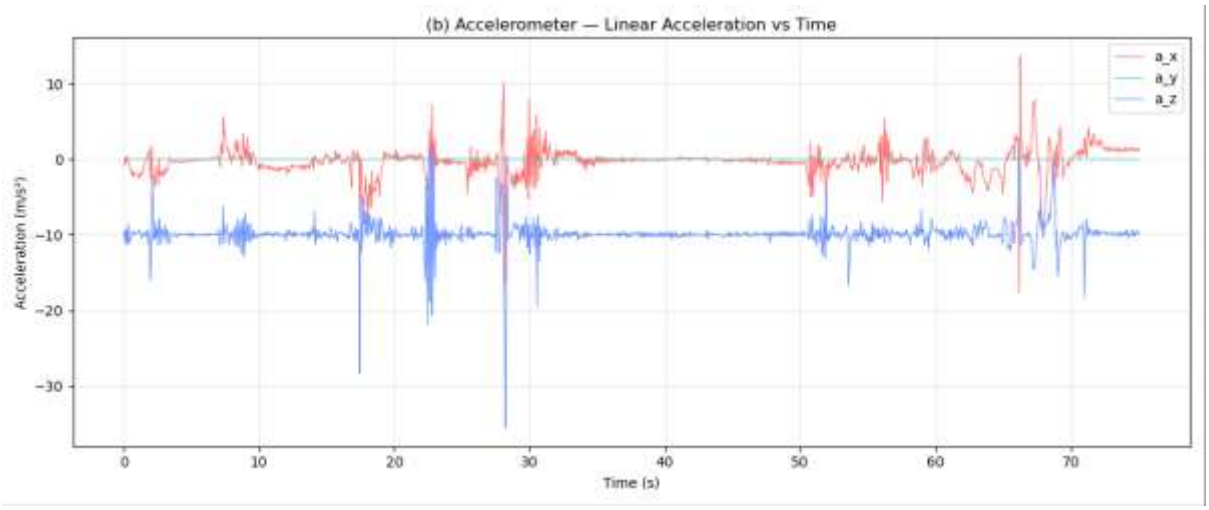


Fig10: Acceleration (X,Y,Z) vs time of moving IMU dataset

Magnetic Data

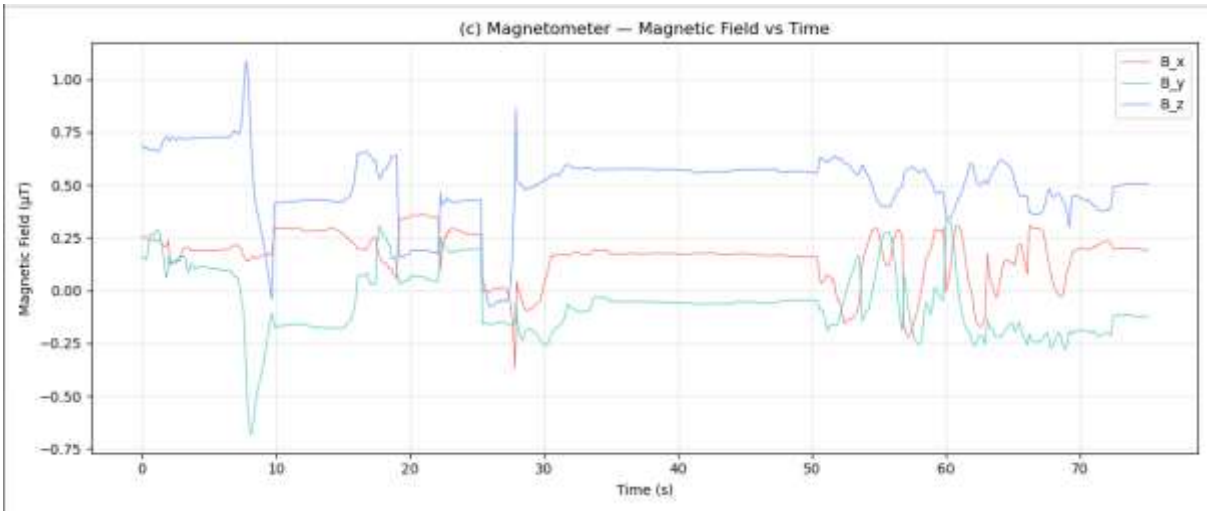


Fig11: Magnetic Field (X,Y,Z) vs time of moving IMU dataset

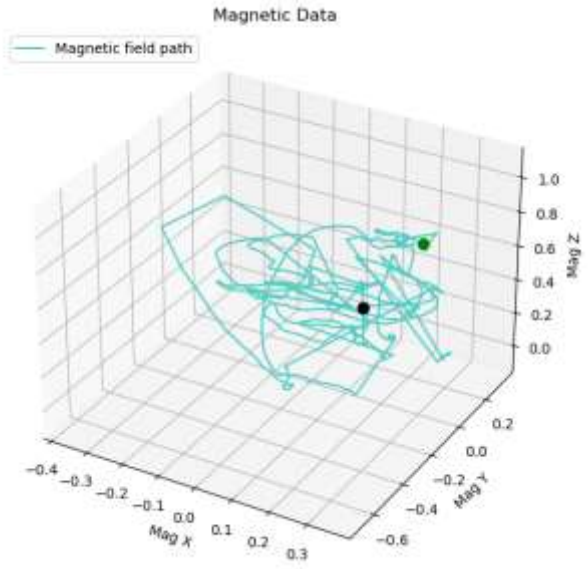


Fig12: Magnetic Field(X,Y,Z) of moving IMU dataset

Gyro Data

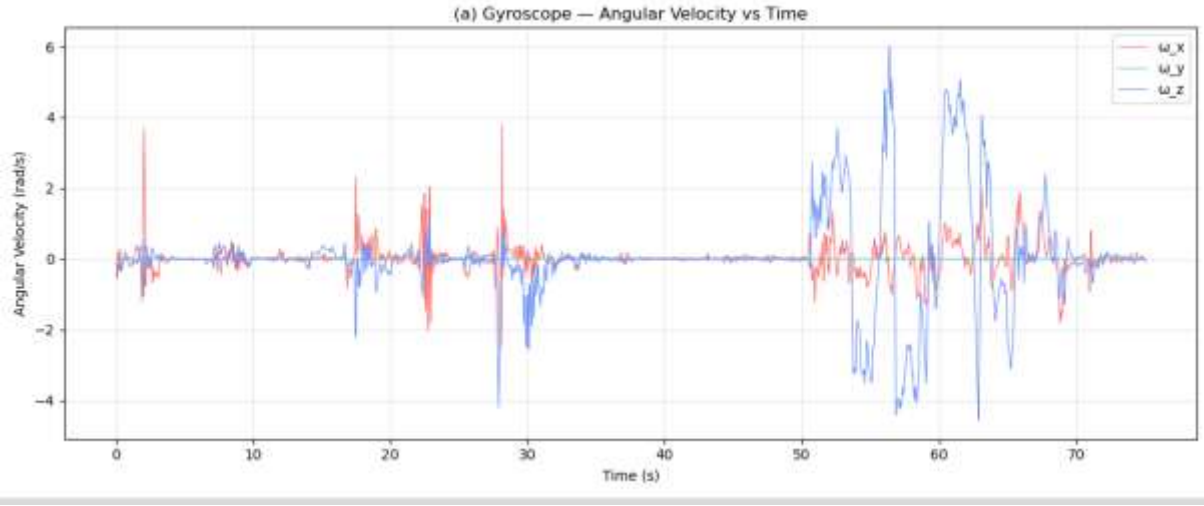


Fig13: Angular rate (X,Y,Z) vs time of moving IMU dataset

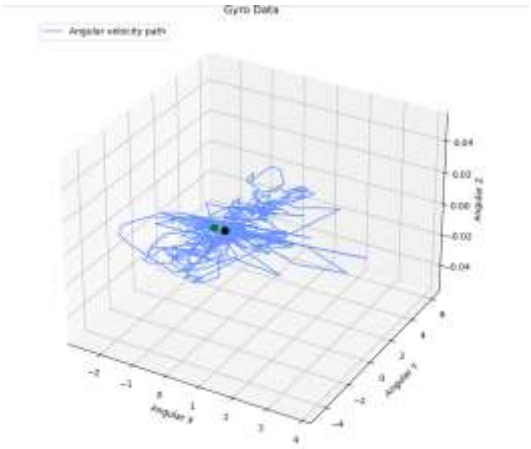


Fig134: Angular rate (X,Y,Z) of moving IMU dataset

This graphs describe the motion of acceleration in X, Y and rotation motion.

The Screengrabs and time series plots of acceleration, angular rate are as follows:

Acceleration in negative Y direction:

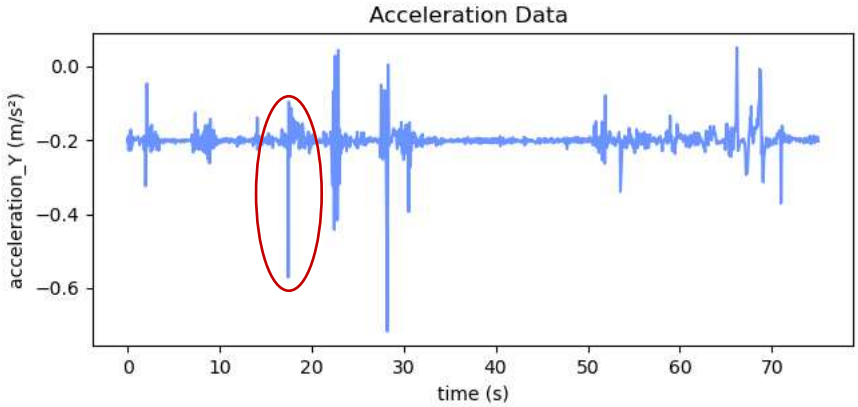


Fig14: Acceleration(Y) vs time of moving IMU dataset

So the movement recording the acceleration in Y direction is observed at 12s in video recording whereas it is observed at ~16s in the data collected and at the exact moment there is some negative acceleration observed in X-direction as well.

The sensor is hold in straight position and dragged towards the left indicating negative acceleration in Y direction. The screen grab for this motion is:



Fig14: Picture showing, accelerating the IMU in X direction

Moving back and forth about X axis: (Acceleration about X axis)

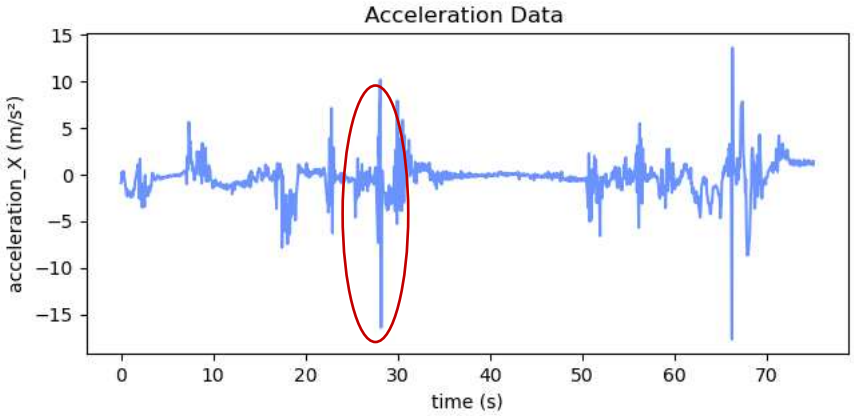


Fig15: Acceleration(X) vs time of moving IMU dataset

Similarly, the marked peak is constituting 2 motions, acceleration in positive X direction, followed by acceleration in negative X direction which is the intended action and can be seen in the video and hence the peak is observed in both directions. This is recorded in the video from 25-28s (almost 3s) interval.

The sensor is held in a position wherein the X-axis is pointing towards the camera and moved back and forth about a line with the same sensor orientation .The screengrab for this motion is :

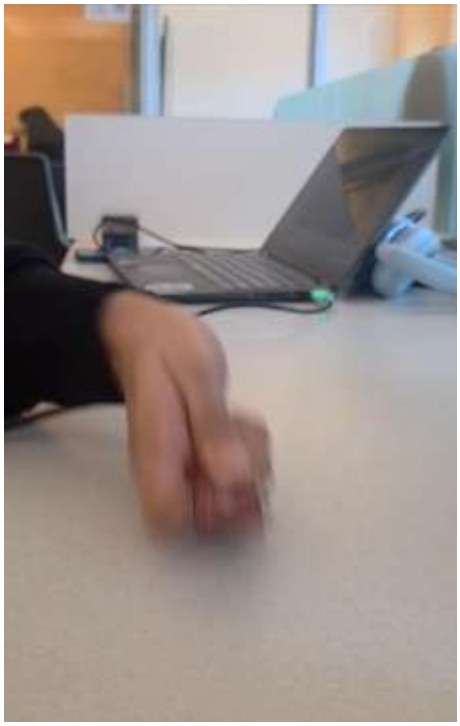


Fig16: Picture showing the acceleration of IMU in X direction

Rotation about Z-axis:

The next motion observed was the rotation about z-axis, which can be visualized as yaw vs time, and also as circular trajectory in X-Y plane when plotted as angular_x(rad/s) vs angular_y(rad/s)

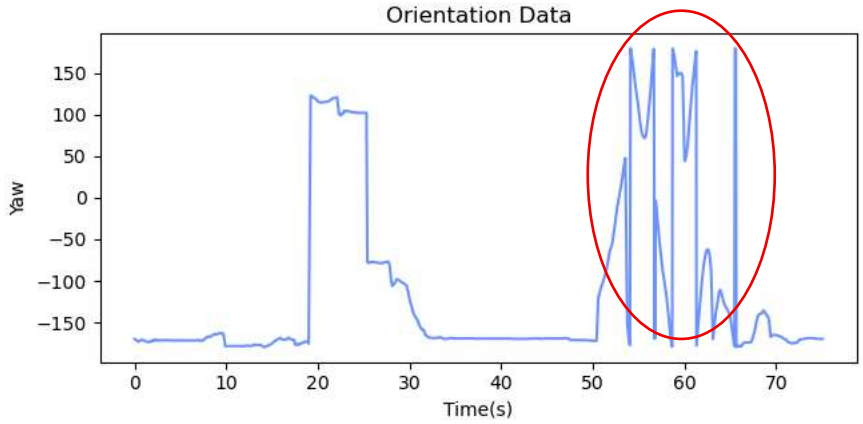


Fig17: Yaw (Z) vs time of moving IMU dataset

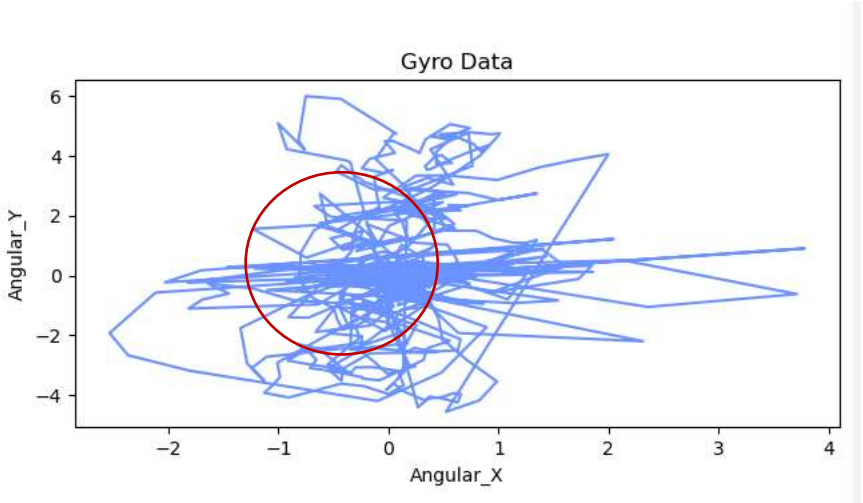


Fig18: Angular rate (X) vs Angular_rate(Y) of moving IMU dataset showing circular motion about Z

The rotation motion is recorded starting from 53s and is stopped around 63s as seen in the above plot. The timestamps align with those in the video recording as well.

The screen grab of this motion is :

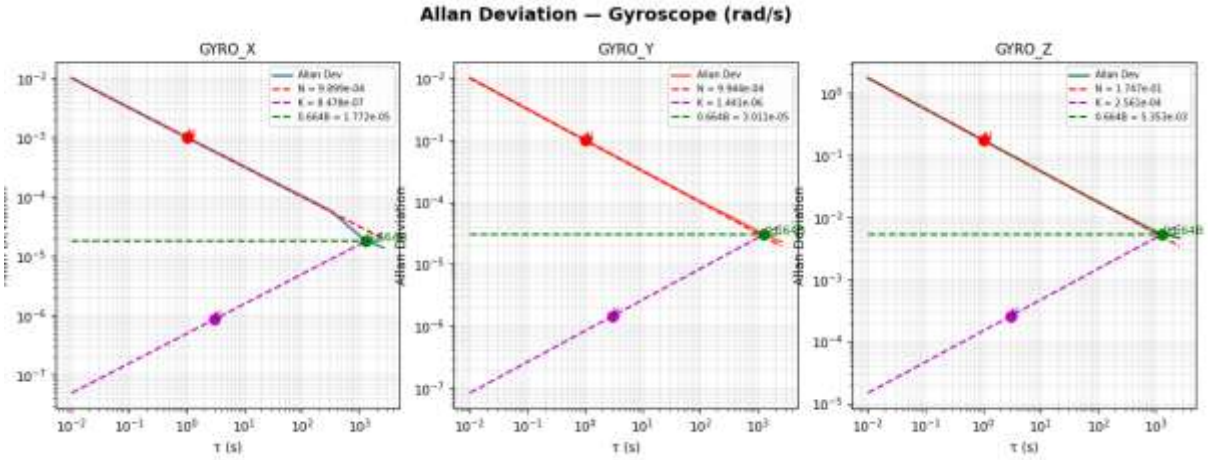


Fig19: Picture showing the rotation of IMU about Z axis. (circular motion)

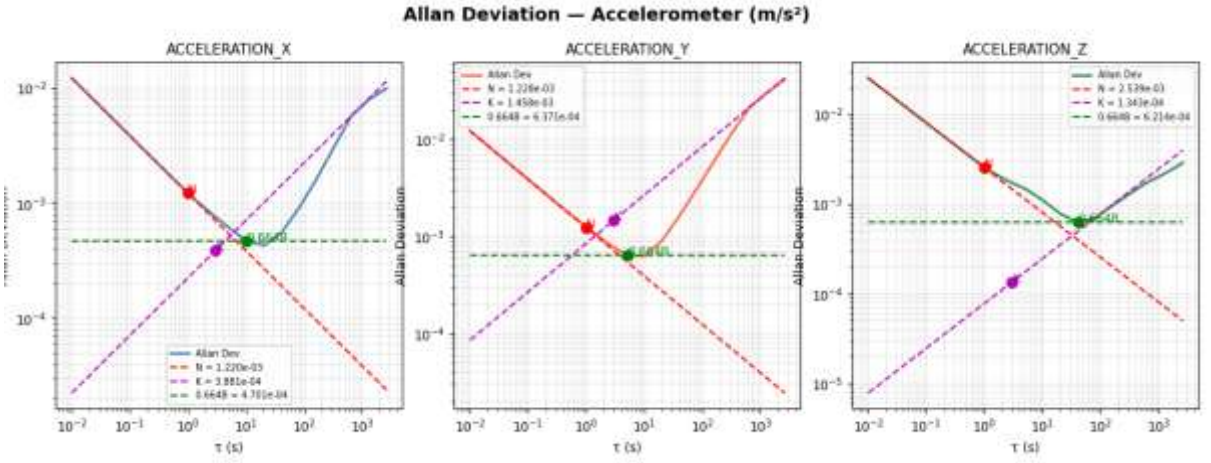
Allan Variance Analysis:

Using the process described on the Mathworks page, for noise parameter identification, these are the allan deviation plots with the values of Rate random walk, Bias Instability and Angle random walk .

The dataset I have analysed is : 5_hour_imu_data



Parameters	Gyro_X	Gyro_Y	Gyro_Z
Rate random walk (K)	8.47e-7	1.441e-6	2.561e-4
Angle random walk(N)	9.899e-4	9.944e-4	1.747e-1
Bias Instability (B)	1.772e-5	3.011e-5	5.353e-3



Parameters	Acc_X	Acc_Y	Acc_Z
Rate random walk (K)	3.881e-4	1.458e-3	1.343e-4
Angle random walk(N)	1.220e-3	1.228e-3	2.539e-1
Bias Instability (B)	4.701e-4	6.371e-4	6.214e-4

Inference:

From the Allan Deviation of gyroscope data, it is observed that the rate random walk or the K noise parameter is of very low order, indicating that there is very less long term drift due to accumulated rate noise for both X and Y axes and of order -4 for the Z axis.

For the X and Y axes, the angle random walk(N) is much less than for the Z axis, indicating that Z-axis is bound to have higher short term noise in angular velocity.

The Bias instability for X and Y axes, is much lower than that of Z-axis indicating that the constant offset affecting long term measurements for Z-axis is would be on the higher side always.

The same results are obtained for the acceleration data as well, thus we conclude that Z-axis were affected by certain factors like environmental etc.

My team-mates also did allan deviation for different datasets provided and had the ranges of noise parameters close to my analysis. Their Z-axis values were almost in the same range as other axes .

Questions:

1) What can you say about the noise in your short stationary data?
From the short duration dataset, we can interpret that the gyroscope data is mostly stable but reports a few variations along X,Y and Z direction which indicates the presence of sensor noise although the IMU is stationary. These variations are caused by bias instability and high frequency measurements.
The accelerometer data is mostly stable with only few noisy measurements specifically along the Z-axis which can be accounted for
The magnetometer data also shows small variations around a fixed value, which is expected when the sensor orientation does not change. Minor deviations especially along the Z-axis may be caused by environmental magnetic disturbances or electronic noise.
The orientation (roll, pitch, yaw) plot shows very small changes over time, confirming that the sensor remained stationary. The small variations in the Euler angles indicate the effect of noise propagation from the gyroscope, accelerometer, and magnetometer through the sensor fusion algorithm inside the VN-100.
The noise appears to be random rather than systematic which suggest that the dominant sources are bias instability and environmental disturbances rather than actual motion
2) What was the median, mean, and distribution?

Mean Yaw = -171.43 degrees

Median Yaw = -171.56 degrees

Mean Pitch = 1.39 degrees

Median Pitch = 1.13 degrees

Mean Roll = 0.13 degrees

Median Roll = 0.13 degrees

From this data I can conclude that, the IMU is more sensitive to Yaw and less sensitive to Pitch and Roll.

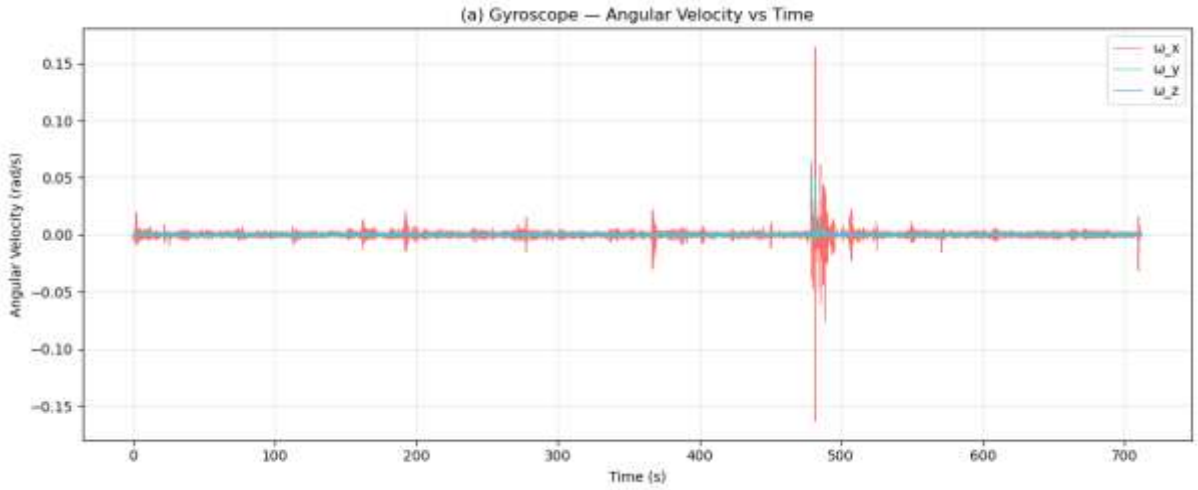
Yaw distribution: Roughly bimodal- two peaks observed around -172 and -170.4 and no peak around the mean with a spread of ~3 degrees .

Roll distribution: Single peak distribution with spread of about 0.2 degrees with mean and median at 0.13 degrees.

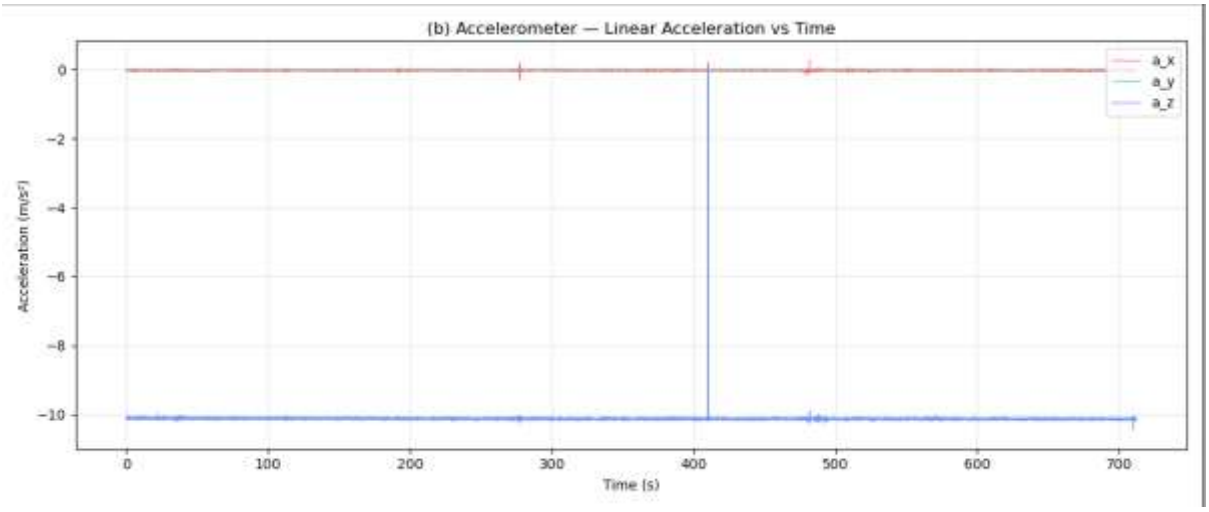
Pitch distribution: Single peak observed around the mean 1.39 degree, very certain/ tight distribution.

3) How do your measurements compare to the performance listed in VN -100 datasheet?

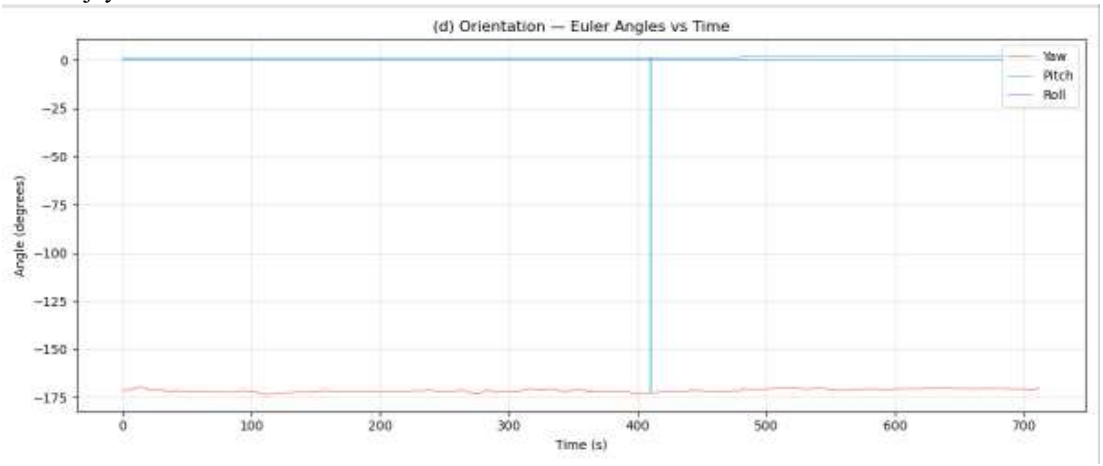
The noise density in gyroscope measurements is 0.0035 °/s/√Hz which is way too low(e-5) in radians given 1Hz frequency. Our data has much higher noise than this causing peaks at random times especially in the X (~0.15 rad/sec)and Y(~0.05 rad/sec) direction even if the sensor is stationary, indicating that the gyroscope measurements are more sensitive to environmental factors in X and Y direction rather than Z direction which is very stable.



For the acceleration data, the noise density is ~0.14 mg/√Hz which is 0.0013734 m/s^2 which is very small as compared to what is observed in our data, given frequency 1Hz. In our dataset, most data is stable but with high noise than specified, however acceleration_z is observed at a constant value of -10 m/s^2 because of gravity.



Orientation data is mostly stable with likely ranges of 0.5 degrees RMS with pitch reporting a sudden spike, can be due to some jerk applied.



4) If the performance in your dataset was worse than the datasheet, what sources of noise do you think were present?

In our dataset, the observed noise exceeds the VN-100 datasheet specifications for both gyroscope and accelerometer measurements. The gyroscope shows peaks of ~0.15 rad/s in X and ~0.05 rad/s in Y even when stationary, while Z remains very stable. This suggests that the X and Y axes are more sensitive to environmental vibrations or some jerks to the cable as the sensor was not mounted but kept unlike the datasheet. The accelerometer shows higher than expected noise, likely due to the sensor not mounted but kept still and ambient disturbances, though the constant -10 m/s² on the Z-axis reflects the influence of gravity, which is expected. Orientation spikes, especially in pitch, indicate occasional jerks or small shocks, possibly from handling or external movement. Overall, the main noise sources are environmental vibrations, mounting instability, and sudden motion.

5) How did you align the video and the VectorNav action?

Negative Y acceleration**:** In the video, this motion occurs at 12 s, but in the IMU data it appears at approximately 16 s**.** At that same IMU timestamp, a small negative acceleration is also observed in the X-direction, indicating motion along X-axis as was intended

Positive and negative X accelerations: A peak in the IMU data corresponds to two consecutive X-direction motions, first positive, then negative. This motion occurs between 25–28 s in the video, which matches the IMU peak observed in both X and Y axes due to simultaneous effects. But the motion was intended to be acceleration along X-axis, the movement might have induced some movement along the Y-axis as well.

Z-axis rotation**:** The yaw rotation begins around 53 s and ends at 63 s, observable in both the video and IMU data. When plotting Angular X vs Angular Y, this rotation forms a circular trajectory in the X-Y plane, further confirming alignment.

By using these correlated events, the time offset between video and sensor data (~5 s) was determined and corrected, ensuring that all motions captured by the IMU align accurately with the visual evidence in the video.

6) Did IMU data correlate well with the video of the actual motion?

Yes, it did. We were able to record the data and visualize it using various IMU quantities to correspond it to the motion.

7) Can you make any inferences about the conditions under which your long-duration data was taken? For example, was there any periodicity to the noise indicating the presence of a machine operating nearby

   From the Allan Deviation charts, we can infer that, the long-duration data was collected in a mostly stable environment, free from periodic mechanical disturbances. Short-term noise in the X and Y axes indicates sensitivity to minor environmental vibrations or mounting micro-movements, while Z-axis data remains very stable. Long-term bias is low, and no rate random walk is observed, confirming that the sensor bias was stable over time. The accelerometer Y-axis reflects gravity, and X/Z axes show mostly random noise with minimal drift. Overall, the Allan variance analysis confirms that the data conditions were suitable for assessing intrinsic sensor noise.

8) What do your results tell you about the conditions you should measure Allan variance under to understand the best sensor performance possible?

   Stable mounting: Place the sensor on a rigid, vibration-free platform to eliminate mounting-induced micro-movements.

   Stationary sensor: Keep the sensor fixed in orientation; any motion will introduce additional noise and bias drift that mask the true sensor performance.

   Minimal environmental disturbances :Perform measurements in a quiet environment with no nearby machinery, motors, or fans that could induce vibrations

   Place the sensor in such a place that has constant temperature and air currents aren’t that heavy.

   Long-duration data collection: Record for sufficient time (minutes to hours) to capture long-term bias instability and low-frequency noise.

   Short averaging times will reveal white noise (angle random walk), while long averaging times reveal bias instability and rate random walk.

   Axis orientation consideration: Ensure the axes are oriented consistently and gravity effects are accounted for

9) If you were trying to characterize a new sensor with unknown performance, briefly (a few sentences) explain what, where, and how you would measure to understand sources of noise in the sensor and environment, and the performance you could expect under its operating conditions.

   To characterize a new sensor, I would first place it on a rigid, stationary platform and record long-duration stationary data to estimate baseline noise, bias instability, and drift. Next, I would perform controlled motions (rotations and accelerations) along each axis to measure sensitivity, scale factors, and axis coupling. Throughout the process, I would monitor environmental factors such as vibrations, temperature changes, and electromagnetic interference to separate intrinsic sensor noise from external disturbances. By comparing stationary and dynamic data, I could identify dominant noise sources and estimate the expected performance of the sensor under typical operating conditions